CampusCart

System Analysis & Design

Technical Report

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Introduction

Background & Business Problem:

College students often face difficulties in acquiring affordable essentials such as textbooks, electronics, kitchenware, dorm furniture, and more. While informal solutions like group chats, campus bulletin boards, and social media groups exist, they suffer from significant limitations — lack of structure, poor discoverability, absence of trust mechanisms, and no formal moderation. These unstructured exchanges often lead to frustration, wasted time, and missed opportunities.

There is a clear opportunity to address this problem by building a centralized, campus-exclusive platform that enables students to exchange second-hand items in a secure, trusted, and scalable environment. Such a solution would not only support students' financial well-being but also foster sustainability through the reuse of goods.

Trigger Point:

Respond to an Opportunity
CampusCart is designed to capture an untapped opportunity in campus ecosystems: promoting sustainability, affordability, and community engagement. By providing a dedicated space for buying and selling second-hand items, the platform helps reduce financial strain on students while simultaneously encouraging the circular use of resources, which contributes to campus-wide environmental goals. The platform also promotes interaction and connection among students, building a stronger sense of community.

Stakeholders and Their Interests:

- **Students (Buyers/Sellers):**
  Seek a trustworthy, user-friendly platform that allows them to buy, sell, or exchange items quickly, safely, and affordably with fellow students.

- **Facilities Team:**
  Interested in providing support and interventions when disputes or operational issues arise in the transaction process, ensuring a smooth and safe experience for all users.

- **University Clubs/Departments:**
  Eager to leverage in-app advertisement space to promote club activities, departmental programs, and campus events, thereby increasing their visibility

and engagement with the student body.

- **University Management:**
  Aims to enhance campus life by offering value-added services to students, promoting sustainable practices through reuse, and fostering a sense of community through peer-to-peer interaction.

Scope of the System:

The CampusCart platform will focus on enabling **peer-to-peer item exchanges** through core functionalities such as:

- **Account Creation with Campus Email Verification:**
  Ensures that only verified students and authorized users can participate in the marketplace.

- **Item Listing and Categorization:**
  Allows users to create detailed listings with photos, item descriptions, and categorized tags for easy browsing.

- **Search and Filter Capabilities:**
  Provides robust search and filter options based on item categories, price ranges, and keywords to help users find relevant items quickly.

- **In-App Messaging:**
  Enables seamless, secure communication between buyers and sellers for negotiation and transaction coordination.

- **Moderation and Safety:**
  Incorporates moderation mechanisms to maintain listing quality, prevent inappropriate content, and ensure a safe user environment.

The app will be designed with a **minimalist, intuitive interface**, making it cost-effective to develop and easy for students to adopt. The lightweight design also ensures smooth performance across a wide range of mobile devices and platforms.

System Vision Document

Problem Description:
Currently, there is no formal or unified digital platform that facilitates safe and efficient item exchange among students on campus. Students rely on informal channels such as bulletin boards, social media groups, or word of mouth, which often lack verification, transparency, and visibility. This fragmented approach leads to low trust among users, limited adoption, inefficiencies in finding or selling items, and missed opportunities for promoting sustainability through reuse. There is a clear need for a dedicated, trusted platform that can streamline campus-based item exchange while fostering community engagement.

System Capabilities:

- **Verified Student Logins**
  Ensures only legitimate campus users (students, faculty, staff) can access the platform by integrating with campus authentication systems.

- **Item Listing with Photos and Categories**
  Allows users to create detailed listings of items for sale or exchange, including images and category tags for easier browsing.

- **Search and Filter by Category or Price**
  Provides advanced search functionality so users can quickly find items of interest based on categories, price ranges, and keywords.

- **In-App Chat for Negotiation and Coordination**
  Facilitates seamless communication between buyers and sellers through an integrated chat feature, enabling negotiation and transaction coordination within the app.

- **Admin Dashboard for Content Moderation and Dispute Resolution**
  Equips administrators with tools to monitor listings and user activity, manage content quality, and resolve disputes to maintain a safe and trustworthy marketplace.

- **Advertising Space for Campus Clubs and Events**
  Offers space within the platform where campus organizations can promote events, activities, and club initiatives, fostering broader community engagement.

Business Benefits

- **Promotes Sustainability**
  Encourages the reuse of goods and materials, reducing campus waste and contributing to environmental sustainability goals.

- **Supports Financial Inclusion**
  Enables students with limited budgets to access affordable second-hand goods and provides opportunities to earn extra income by selling unused items.

- **Empowers Peer-to-Peer Sharing**
  Builds a stronger sense of community by facilitating direct exchanges between students, fostering collaboration and mutual support.

- **Scalable and Replicable Model**
  Designed as a scalable solution that can easily be adapted and deployed across other campuses or similar communities, providing a valuable tool for NGOs and educational institutions.

Requirements

Current Situation:

At present, students looking to buy and sell items on campus rely heavily on informal channels such as messaging app group chats, physical bulletin boards, word-of-mouth referrals, or ad-hoc social media posts. While these approaches are simple, they are also fragmented, unmoderated, and largely unstructured.

Key limitations of the current situation include:

- **Lack of Verification:** There is no way to ensure that participants in these exchanges are verified members of the campus community, which can lead to trust issues and safety concerns.

- **Limited Visibility and Reach:** Items often remain unseen by many potential buyers due to the fragmented nature of these channels.

- **No Formal Support or Moderation:** There is no structured system for dispute resolution, moderation of inappropriate content, or monitoring of fraudulent activities.

- **Inefficient Discovery:** Without structured listings and search capabilities, it is difficult for students to discover items that meet their needs.

A **dedicated, centralized campus-exclusive platform** would address these shortcomings by enabling secure, trusted, and efficient peer-to-peer item exchange — ultimately enhancing both the user experience and community trust.

Requirements for the Information System:

- **Verified User Authentication:**
  The system must authenticate users via their campus-issued email accounts to ensure only authorized members of the campus community can access the platform.

- **Item Listings:**
  The system must allow users to create item listings that include:

  - High-quality **item photos**

    - Detailed **item descriptions**
    - Relevant **categories** (e.g., Books, Electronics, Furniture, Clothing, etc.)

- **Search and Filter Functionality:**
  The system must provide a robust search capability with filters to enable users to:
    - Search for items by **category**, **price range**, and **keywords**
    - Sort results based on relevance, date of posting, and price

- **In-App Chat Feature:**
  The system must include a secure **in-app chat** feature that allows buyers and sellers to:
    - Negotiate item prices and terms of exchange
    - Coordinate pickup logistics and meeting times
    - Maintain privacy by not exposing personal contact information

Individuals to Interview:

1. Students (Buyers & Sellers)
2. Facilities Team Member (Housing Department)
3. University Club Representative

Interview Objectives & Questions

1. Students

Objective: Understand user expectations, challenges in current item exchanges, and desired app features.

Sample Questions:

- How do you currently buy/sell second-hand items on campus?
- What challenges do you face with current methods?
- What features would be most useful in an app for item exchange?
- How concerned are you about trust and safety while transacting with peers?
- Would you prefer meeting in a common campus location for exchanges? Why or why not?
- Have you ever faced any issues or disputes while exchanging items with other students?
- How likely are you to use a campus-exclusive app if it verifies users through university login?

2. Facilities Team

Objective: Understand the team's role in handling disputes or issues that arise during item exchanges between students.

Sample Questions:

- Have there been past instances where your team had to intervene in student-to-student transactions?
- What kind of disputes or issues are you typically called upon to resolve?
- What support or resources would your team need to handle potential conflicts related to item exchanges?
- Are there any policies or protocols in place for addressing such student interactions on campus?

3. University Club Representative

Objective: Explore advertising needs and platform collaboration for event promotions.

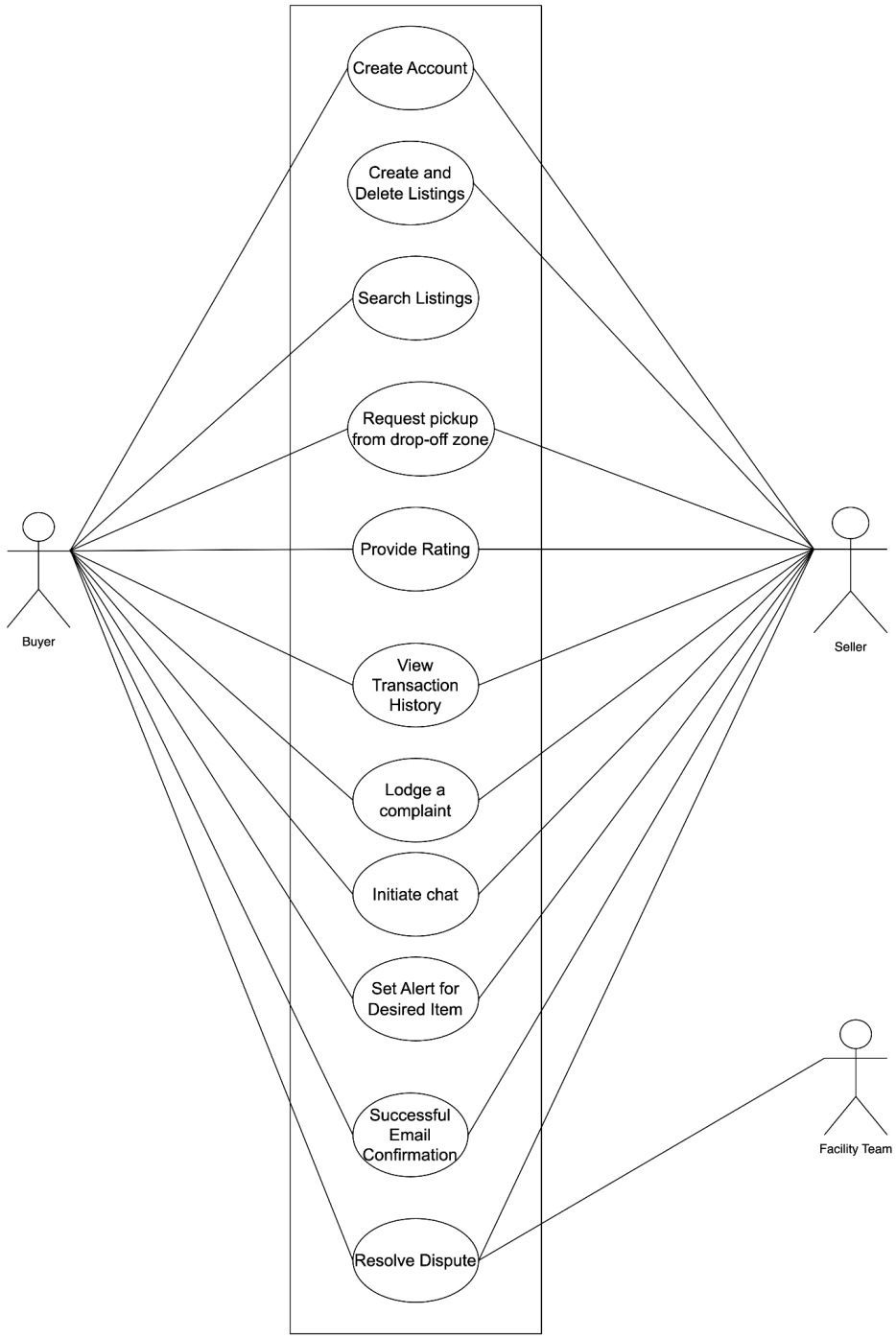
Sample Questions:

- Would your club be interested in promoting events through this app?
- What kind of visibility or engagement are you looking for?
- Would you consider paying a small fee for promotion?
- What type of content or formats (banners, event listings, pop-ups) would best suit your promotions?
- How frequently would your club be interested in advertising events on the platform?
- Would you prefer targeting specific student groups or keeping promotions visible to all users?

UML Diagrams

Use Case Diagram

The Use Case Diagram provides a high-level view of the system's functionality by illustrating the interactions between users (Buyers and Sellers) and the system's core services.



Use Case Descriptions

Use Case Name: Request Item Pickup From Drop-Off Zone

This use case describes how a Buyer can request the pickup of an item from a designated drop-off zone within the system.

Usecase Name:	Request item pickup from drop-off zone	
Scenario:	A buyer requests item pickup from a designated drop-off zone on campus.	
Triggering Event	Buyer and seller agree on the sale, and buyer opts for pickup at a drop-off zone.	
Breif Description:	After submitting the item pick up request, the buyer selects a preferred drop-off location and time slot. The system notifies the seller and facilities team. The item is then made available for pickup from the chosen zone.	
Actors	Buyer, Seller	
Related Use Case	Rate Seller Successful Swap Email Confirmation	
Stakeholders	Facilities Team Campus Security	
Preconditions	Item pick up request submitted. Drop-off service must be enabled Drop-off service must be enabled	
Postconditions	Pickup request is recorded Drop-off slot is booked Seller & Buyer receive PIN Item is marked as “awaiting pickup”	
Flow of Activities	Actor	System
	1. Buyer submits the item pickup request 2. Buyer selects preferred zone and time 3.Seller accepts or requests reschedule --> If Accepted --> If not Accepted(Reschedule) 4.1 Seller selects new preferred zone and time 5. Buyer accepts or reschedules again	1.1 System prompts for drop-off location and time 2.1 System notifies seller of buyer’s preferred zone and time 2.2 Seller receives the notification *(Decision: Accepted?)* 3.1 System confirms pickup request 3.2 System generates PIN 3.3 System notifies both buyer and seller with the PIN 3.4 System updates item status to "awaiting pickup" 4.1.1 System notifies buyer of seller’s new preferred zone and time *(Cycle may repeat until confirmed)*
Exception Conditions:	1.1 Seller doesn’t respond in time 2.1 System is not available 3.1 Item is still listed while confirming the booking	

Use Case Name: Set Alert for Desired Item

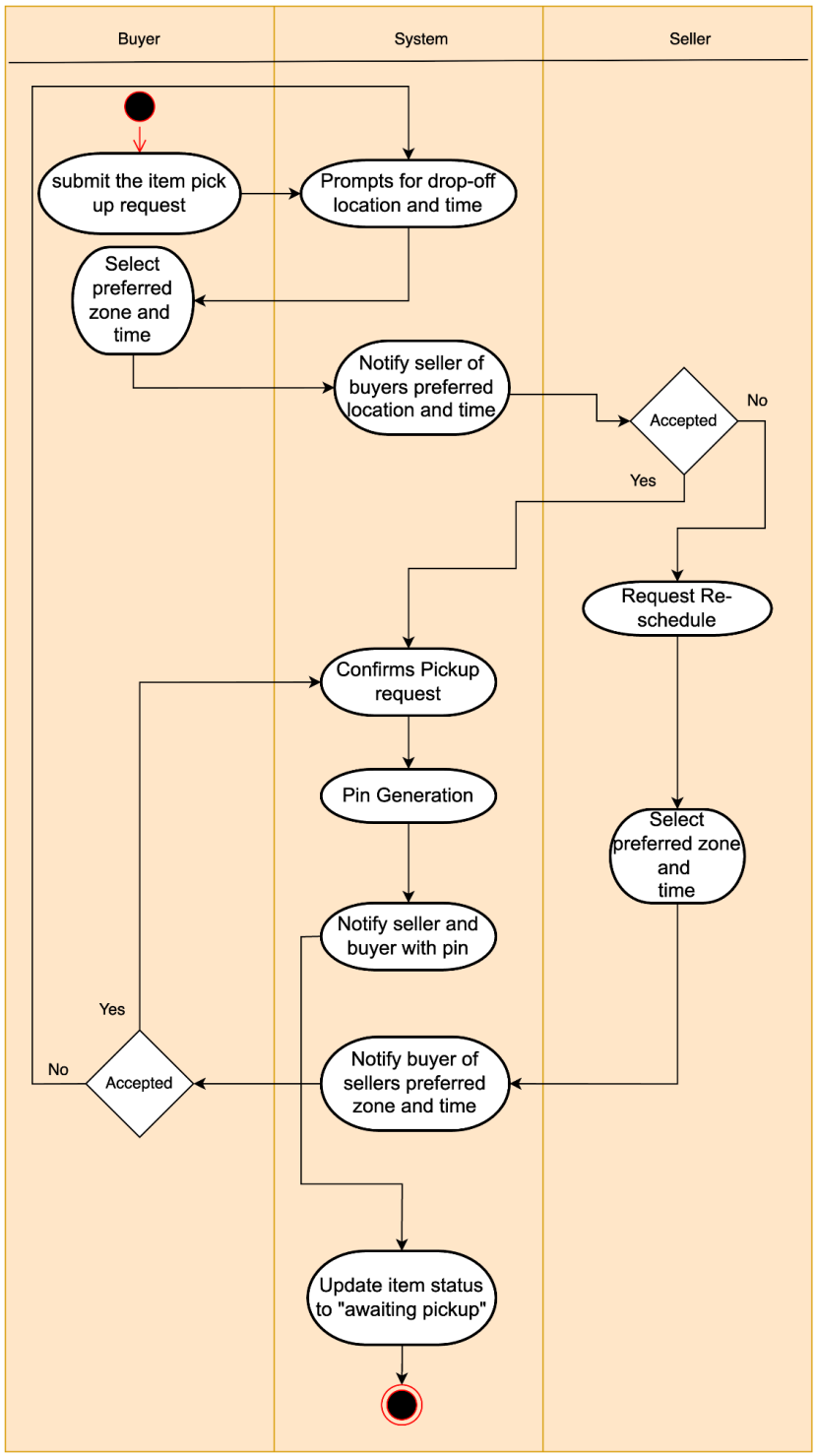
This use case describes how a Buyer can set an alert for a desired item, so they are notified when matching listings become available in the system.

Usecase Name:	Set Alert for Desired Item	
Scenario:	A buyer wants a specific item but it's currently not listed. The buyer sets an alert based on item keywords or category.	
Triggering Event	Buyer cannot find a desired item during search and opts to be notified when it becomes available.	
Breif Description:	The buyer creates an alert specifying a category or keywords (e.g., “Table,” “CS101 textbook”). When a seller later posts an item that matches the alert criteria, both the buyer and the seller receive notifications – enabling faster communication and potential transaction.	
Actors	Buyer, Seller	
Related Use Case	Search Listings Create Listing	
Stakeholders	Buyers Sellers	
Preconditions	Buyer must be logged in Alert system must be functional Alerts must be enabled for both parties	
Postconditions	Alert is saved in the system If a matching item is posted, buyer and seller are alerted Optionally, buyer can initiate chat or proposal	
Flow of Activities	Actor	System
	1. Buyer searches for an item and finds no matching results 2. Buyer clicks “Set Alert” and defines criteria (e.g., category, keyword) 3. Later, Seller posts a new listing	1.1 System prompts “Set Alert” option 2.1 System saves alert configuration and enables background monitoring 3.1 System matches new listing with stored alerts 3.2 If a match is found, system sends alert to Buyer (with listing link) & seller with user profile
Exception Conditions:	1.1 Buyer enters invalid or vague keywords 3.1 No alerts match the new listing 3.2 Alerts failed due to system issue or user settings	

Activity Diagrams

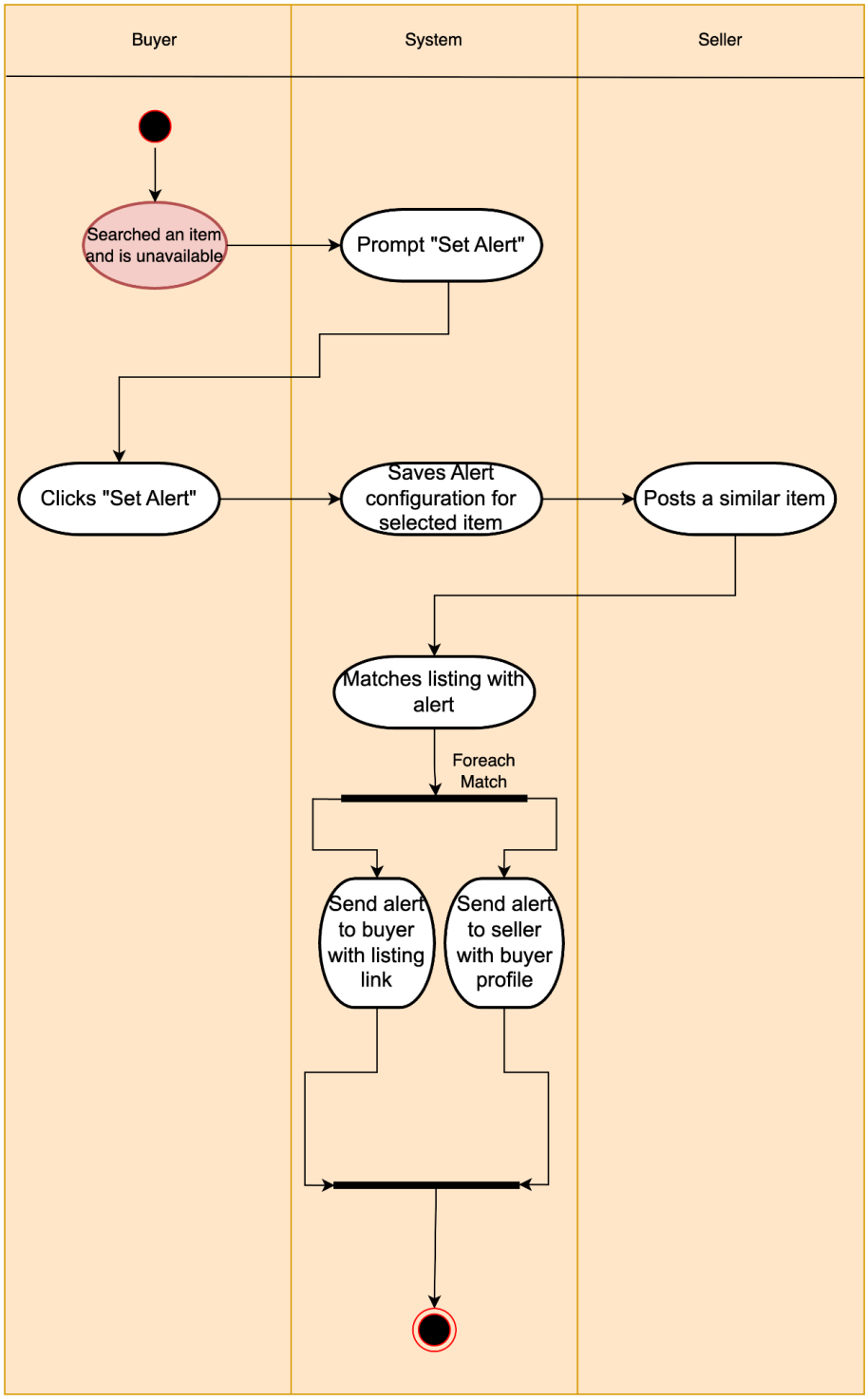
Activity Diagram: Request Item Pickup From Drop-Off Zone

This activity diagram illustrates the step-by-step flow of actions involved when a buyer requests the pickup of an item from a designated drop-off zone.



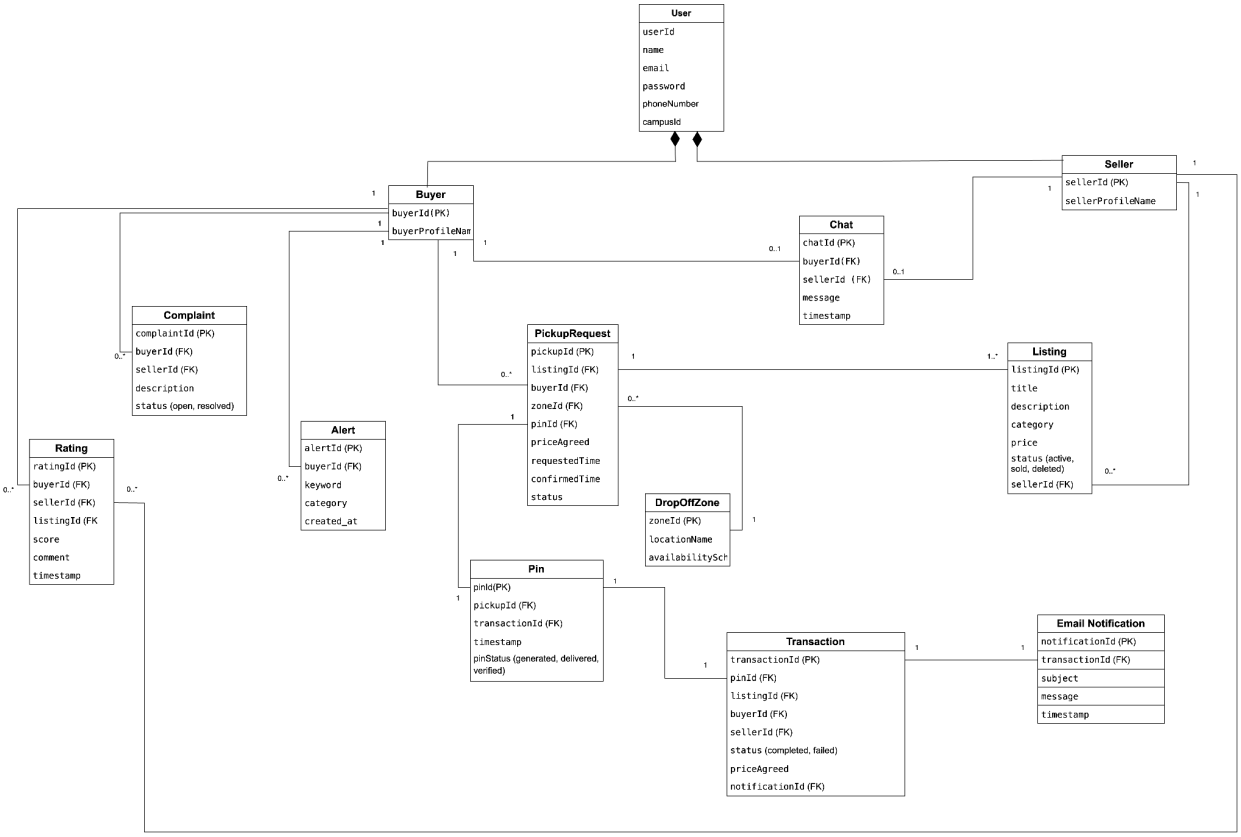
Activity Diagram: Set Alert for Desired Item

This activity diagram illustrates the sequence of actions a Buyer performs to set an alert for a desired item.



Domain Class Diagram

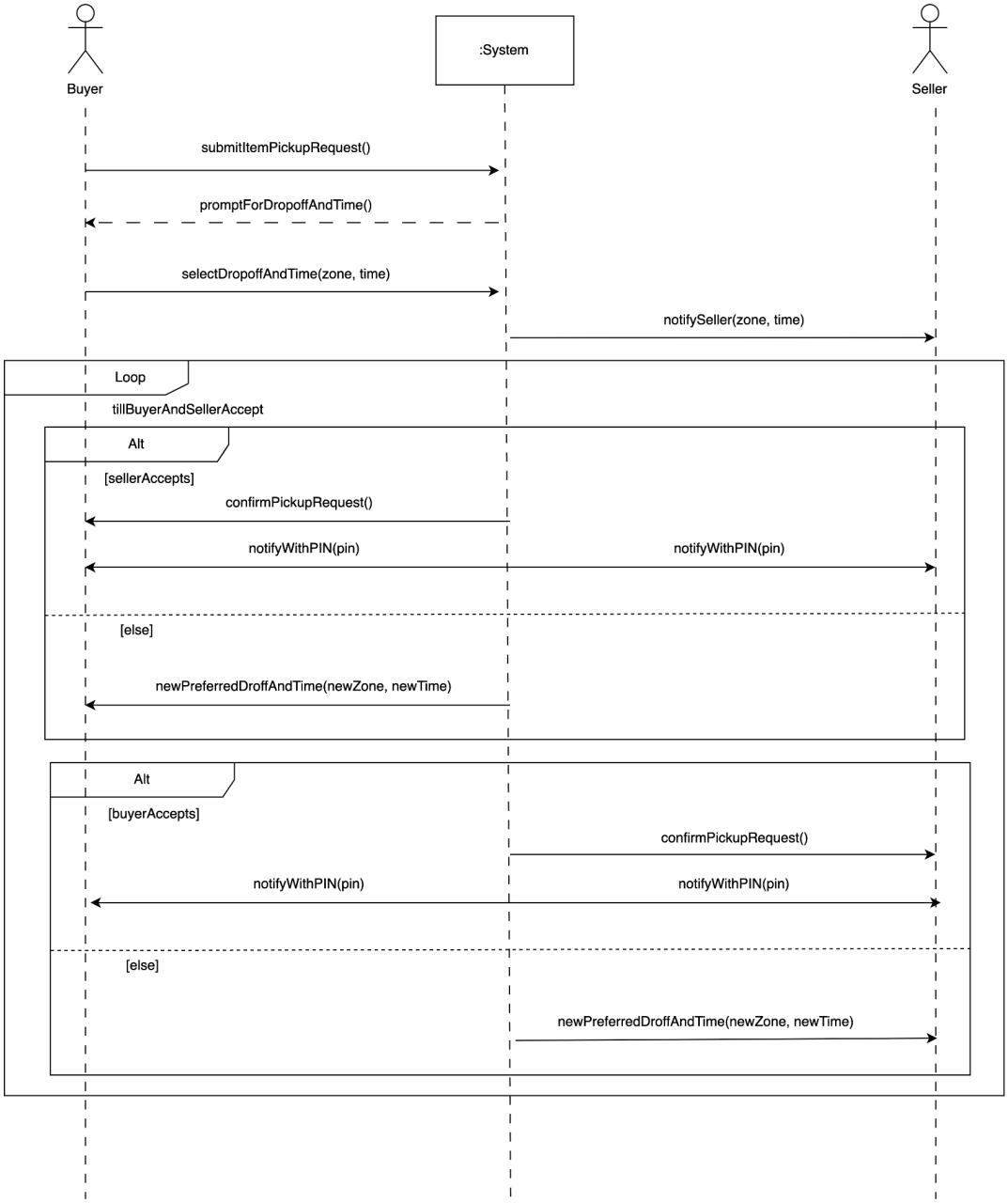
The domain class diagram represents the key entities of the system and their relationships, providing a conceptual view of the system's data structure.



System Sequence Diagrams

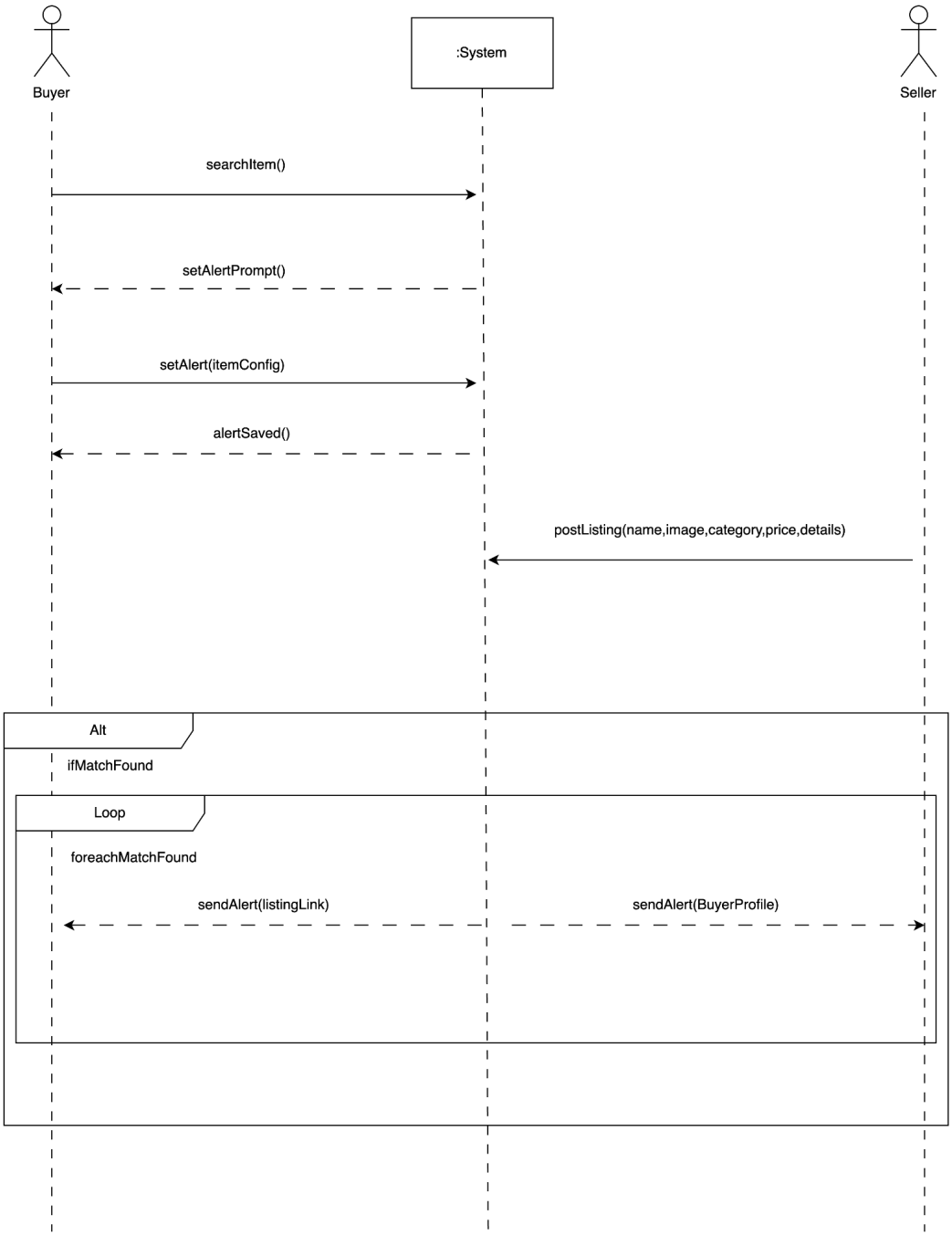
Request Item Pickup From Drop-Off Zone

This system sequence diagram depicts the interactions between the buyer, seller, and the system when initiating a request to pick up an item from the drop-off zone.



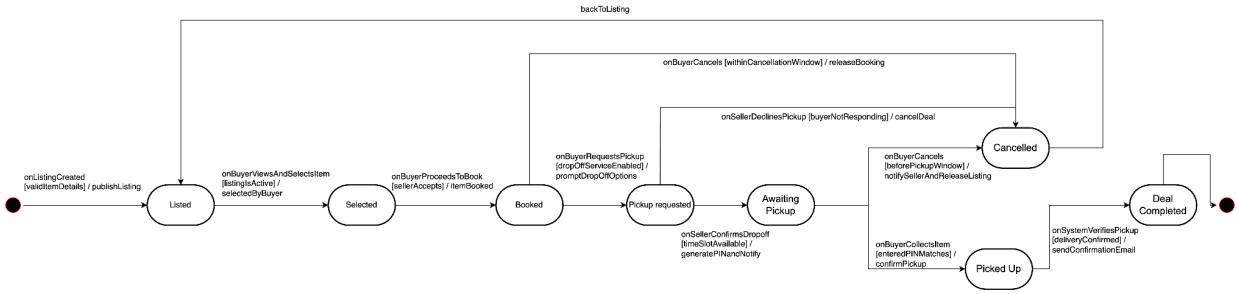
Set Alert for Desired Item

This system sequence diagram illustrates the interactions between the buyer, seller, and the system when setting an alert for a desired item.



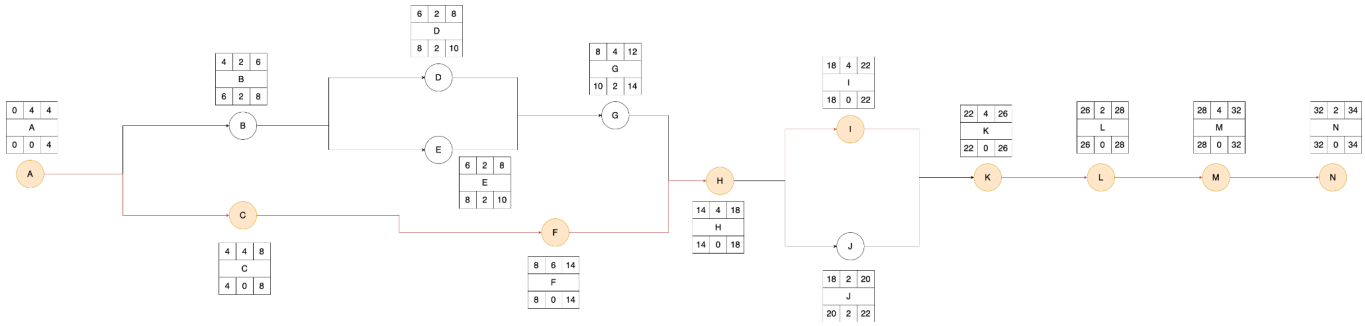
State Diagram

The state diagram illustrates the various states an item can transition through within the system, based on different events and actions.



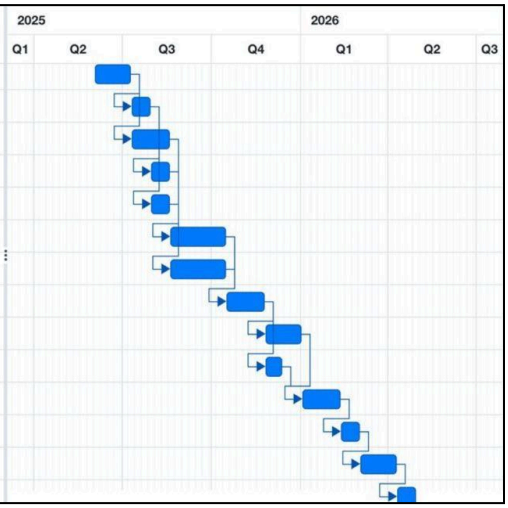
Network Diagram

The network diagram provides an overview of the system's deployment architecture, showing how various components and nodes are connected across the network.



Gantt chart

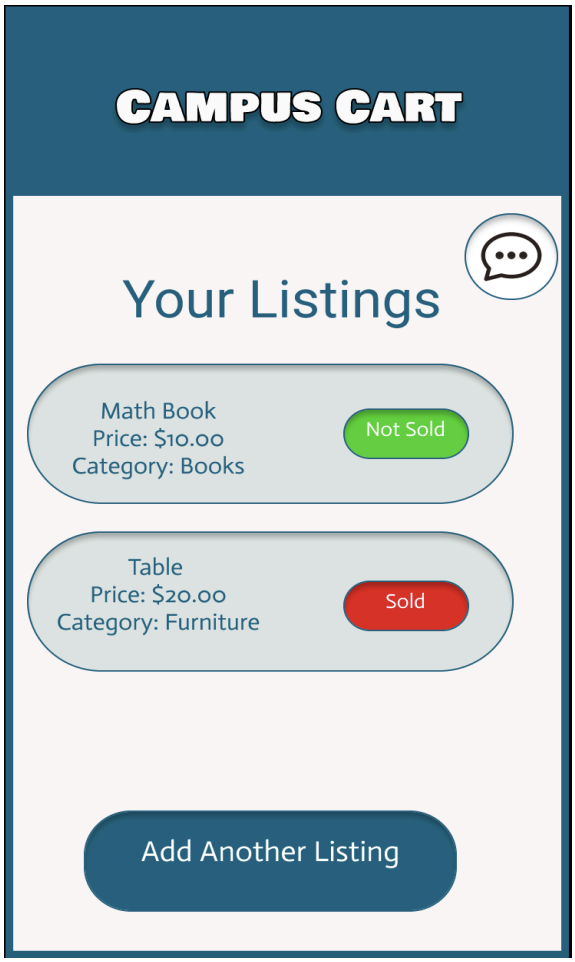
ID	Task Name	Start	End	Duration
1	A - Requirement Gathering & Stakeholder Inte···	2025-06-02	2025-07-09	28 days
2	B - System Architecture & Tech Stack Finaliza···	2025-07-10	2025-07-29	14 days
3	C - UI/UX Mockups & Use Case Finalization	2025-07-10	2025-08-18	28 days
4	D - Database Schema Design	2025-07-30	2025-08-18	14 days
5	E - API Design Specification	2025-07-30	2025-08-18	14 days
6	F - Frontend Interface Development	2025-08-19	2025-10-15	42 days
7	G - Backend Core Development	2025-08-19	2025-10-15	42 days
8	H - Integration of Frontend and Backend	2025-10-16	2025-11-24	28 days
9	I - Security & Privacy Implementation (Login/A···	2025-11-25	2026-01-01	28 days
10	J - Payment Gateway & Transaction Fee Setup	2025-11-25	2025-12-12	14 days
11	K - Feature Testing (Swap Logic, Ratings, Not···	2026-01-02	2026-02-10	28 days
12	L - Admin Panel & Dispute Management	2026-02-11	2026-03-02	14 days
13	M - Final User Testing & Bug Fixing	2026-03-03	2026-04-09	28 days
14	N - Marketing, Deployment & Campus Launch	2026-04-10	2026-04-29	14 days



User Interface Design

The user interface design showcases the visual layout and key interaction elements that enable users to effectively navigate and interact with the system.

Seller's UI



Buyer's UI

Cost-Benefit Analysis

Economic life of the system is estimated to be 5 years

We begin with an initial cost of $24,000 in Year 0. We ensured to keep the cost minimal by utilizing open source technologies and a freelance developer.

The annual operating cost covers hosting and maintenance and is estimated to increase by 10% each year.

On the benefit side, we project growing revenue from subscriptions and ads, starting at $11,000 in Year 1 and reaching nearly $19,500 by Year 5.

As a result:
Our net cash flow turns positive in Year 1, and continues to grow each year.

We applied a 10% discount factor to assess the time value of money.

Year	**0**	**1**	**2**	**3**	**4**	**5**
New system cost	24,000					
Operating cost new system (hosting + maintenance)		2,260	2,486	2,735	3,008	3,309
Total Costs	24,000	2,260	2,486	2,735	3,008	3,309
Revenue: Subscriptions + Ads		11,000	12,650	14,548	16,730	19,239
Total Benefit		11,000	12,650	14,548	16,730	19,239
Net Cash Flow (Benefit - Cost)	-24,000	8,740	10,164	11,813	13,722	15,930
Discount Factor (10%)	1	0.909	0.826	0.751	0.683	0.621

Metric	**Value**
Payback Period (In years)	2.5
NPV	$20,484
IRR	36.11%
Profitability Index (PI)	1.85

Year	1	2	3	4	5
ROI (cumulative)	45.83%	52.71%	60.61%	69.71%	80.16%